THE PICARD STACK IS A GERBE OVER THE PICARD FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1638-1642 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1644-1673, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 11 of queue_moduli.json (pb=1429, stmt 1638-1642, proof 1644-1673).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Author writes 'This follows immediately from ...; here is a detailed explanation' - the detailed explanation (both conditions verified) is the proof and is fully in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0D26: sources/stacks-moduli/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D25: sources/stacks-moduli/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06P2: sources/stacks-moduli/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06P1: sources/stacks-moduli/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06QC: sources/stacks-moduli/pinned-original-source.tex, source lines 1673–1673. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The morphism $\mathcal{P}ic_{X/B} \to \mathrm{Pic}_{X/B}$ turns the Picard stack into a gerbe over the Picard functor.*

Proof. The definition of $\mathcal{P}ic_{X/B} \to \mathrm{Pic}_{X/B}$ being a gerbe is given in Morphisms of Stacks, Definition 06QC, which in turn refers to Stacks, Definition 06P2. To prove it, we will check conditions (2)(a) and (2)(b) of Stacks, Lemma 06P1. This follows immediately from Quot, Lemma 0D26; here is a detailed explanation.

Condition (2)(a). Suppose that $\xi \in \mathrm{Pic}_{X/B}(U)$ for some scheme U over B. Since $\mathrm{Pic}_{X/B}$ is the fppf sheafification of the rule $T \mapsto \mathrm{Pic}(X_T)$ on schemes over B (Quot, Situation 0D25), we see that there exists an fppf covering $\{U_i \to U\}$ such that $\xi|_{U_i}$ corresponds to some invertible module $\mathcal{L}_i$ on X_{U_i}. Then $(U_i \to B, \mathcal{L}_i)$ is an object of $\mathcal{P}ic_{X/B}$ over U_i mapping to $\xi|_{U_i}$.

Condition (2)(b). Suppose that U is a scheme over B and $\mathcal{L}, \mathcal{N}$ are invertible modules on X_U which map to the same element of $\mathrm{Pic}_{X/B}(U)$. Then there exists an fppf covering $\{U_i \to U\}$ such that $\mathcal{L}|_{X_{U_i}}$ is isomorphic to $\mathcal{N}|_{X_{U_i}}$. Thus we find isomorphisms between $(U \to B, \mathcal{L})|_{U_i} \to (U \to B, \mathcal{N})|_{U_i}$ as desired. □